Oh Reum Kwon

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EDUCATION

Georgia Institute of Technology **Aug 2024 - Present**
Master of Science, Computer Science
Relevant Coursework**:** Machine Learning, Computer Vision **Expected: May 2027**

San Jose State University **Aug 2020 - May 2023**
Bachelor of Science, Software Engineering
Relevant Coursework**:** Data Structures & Algorithms, Object Oriented Design, Networks, Operating Systems, DBMS, Security, Enterprise Software

TECHNICAL SKILLS

Quality & Reliability: Statistical Process Control (SPC), Cpk Analysis, Failure Analysis (FA), FMECA, Reliability Testing, Root Cause Analysis
Programming: Python, C++, Java, SQL, MATLAB
ML & Perception: Kalman & Particle Filters, Sensor Fusion (GNSS, IMU, MEMS), Computer Vision
Frameworks & Tools: Tableau, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, OpenCV, Docker, Kubernetes, GCP, Claude Code

WORK EXPERIENCE

Apple | Cupertino, CA **Jan 2026 – Aug 2026**
Display Quality Engineer Intern
- Built predictive models to flag rare downstream test failures from upstream parametric data, targeting units that passed upstream well within spec and were nearly indistinguishable from good units
- Extended analysis beyond traditional SPC and Cpk metrics to establish proactive test parameters and evaluate panel quality
- Built Python and Tableau visualizations comparing passing and failing populations across hundreds of parameters, presenting findings in monthly reviews to engineers and managers

SK Hynix Memory Solutions | San Jose, CA **Nov 2025 – Jan 2026**
SSD Reliability Engineer Intern
- Executed SSD reliability testing, analyzing storage I/O performance and bottlenecks during high-throughput model training
- Implemented GPT-2 from scratch to generate realistic AI training workloads for SSD stress testing
- Developed automated log parsing scripts in Python, reducing manual data review time by 50%

Samsung Semiconductor | San Jose, CA **May 2024 – Aug 2024**
Data Engineer Intern
- Applied ML models to enhance Kalman filter–based algorithms for tunnel detection using GNSS and MEMS data, improving localization accuracy in low-signal environments
- Trained and validated machine learning models in MATLAB to classify tunnel vs. non-tunnel driving conditions with 98% accuracy
- Developed Python automation tools that improved the perception team's workflow efficiency by 30%, by decoding, preprocessing, and visualizing large sensor datasets
- Collaborated with the location team on sensor fusion techniques and data analysis to support advanced positioning and navigation research

PROJECTS

Gauge | Python, FastAPI, PostgreSQL, React **Nov 2026**
- SPC for a simulated display line: $\bar{X}$/R, EWMA and Cpk on luminance, uniformity, ΔE2000 and TFT Vth
- Caught simulated backlight aging as Cpk fell from 1.92 to 0.97 while panels were still in spec
- Cut false alarms across 20 charts by splitting Western Electric rules into alarms and warnings

Computer Vision Object Tracking | Python, OpenCV **Dec 2025**
- Building a robust object-tracking system leveraging Kalman and particle filters for multi-object tracking under noise and occlusion
- Exploring applications in autonomous navigation and perception pipelines